

EchoFinder

Design Document

Version 2 — 0–5 Year Emerging Artist Scope

Find the modern echo of the music you love.

Product type: Spotify-centered music discovery application

Primary value: Translate legacy artists, genres, and scenes into active newer artists

Primary scope: Artists who emerged within the last 0–5 years

Primary modes: Spotify account analysis or manual search by artist / genre / scene

Prepared for Robert McCarn

Executive Summary

EchoFinder is a Spotify-centered music discovery application that helps users find newer, active, decently successful artists who emerged within the last 0–5 years and sound stylistically similar to older favorite artists, bands, genres, or scenes.

The core user problem is that many listeners still love older artists from their formative years, but they want new music that feels connected to those sounds rather than random, overly trendy, or algorithmically disconnected. EchoFinder treats older taste as a useful map and helps users discover modern artists, scenes, and sounds that carry forward the same musical DNA.

EchoFinder supports two primary paths: Spotify Account Mode, where users sign in with Spotify for taste analysis, and Manual Discovery Mode, where users search by artist, genre, scene, mood, or song without signing in.

Core Problem

Many millennial and older Gen Z listeners have strong emotional and stylistic attachments to artists from the 2000s and 2010s. However, those legacy artists may no longer dominate mainstream culture, may not release frequently, or may not sit at the center of current Spotify discovery surfaces.

Standard recommendation tools often answer a broad question: “What else might you like?” EchoFinder answers a more specific question: “What newer artists emerged in the last five years and sound like modern echoes of the artists, genres, and scenes I already love?”

This requires combining several signals that are usually separated across platforms: style similarity, artist recency, current activity, success or traction, scene lineage, and plain-language explanation.

Research and Platform Constraints

Spotify remains the natural user-facing platform for listening, authentication, and playlist creation. However, EchoFinder should not rely on Spotify’s restricted recommendation endpoints as its core engine. Spotify announced in November 2024 that new Web API use cases would no longer have access to several features, including recommendations, related artists, audio features, and audio analysis. Source: [Spotify Developer Blog](#)

Spotify’s February 2026 changelog also removed or deprecated some fields useful for popularity analysis in developer-mode contexts, including artist followers and artist popularity. Source: [Spotify Web API Changelog](#)

EchoFinder should use Spotify as the listening and user-library layer, while building recommendation intelligence from a broader metadata stack: MusicBrainz, Last.fm, release history, tags, playlists, artist descriptions, public activity signals, and user feedback. MusicBrainz offers REST-style music metadata access, and Last.fm provides artist tags and similar-artist endpoints. Sources: [MusicBrainz API](#) and [Last.fm artist.getSimilar](#)

Product Vision

EchoFinder helps users translate older music taste into current music discovery. It should feel like a smart, taste-aware music researcher that can explain how newer artists carry forward the sound, scene, mood, or lyrical DNA of legacy favorites.

Legacy artists + current scenes + recent artist emergence + activity signals + traction signals + explainable similarity = useful modern discovery

Revised Product Scope: 0–5 Year Emerging Artists

The strongest product scope is not “artists formed in the last 5–7 years,” but rather artists who first released, first became active, or meaningfully emerged within the last 0–5 years.

This is better than “formed” because formation dates are often unreliable. Many artists form before their first public release, while others exist informally for years before gaining a meaningful audience. EchoFinder should care most about when the artist became discoverable, active, and relevant to the listener.

- First release within the last 5 years
- First meaningful public activity within the last 5 years
- First playlist, press, or fan traction within the last 5 years
- Breakout period within the last 5 years
- Recent release within the last 12–24 months
- Currently active through releases, touring, playlisting, press, or fan growth

Default MVP Rule

1. Match the user’s legacy artists, selected genre, scene, or Spotify taste profile.
2. Emerge within the last 0–5 years.
3. Show visible activity within the last 24 months.
4. Have enough traction to avoid feeling abandoned or random.
5. Be explainable as a modern echo of the user’s older taste.

Target Users

Primary User

A listener in their late 20s to early 40s who has strong attachment to older artists but wants to discover new music that still feels aligned with their taste.

Example Legacy Artists

- Manchester Orchestra
- Thrice
- The Decemberists
- Brand New
- Death Cab for Cutie
- The Wonder Years
- Bright Eyes
- Modest Mouse
- Coheed and Cambria
- Frightened Rabbit
- The National
- The Menzingers

Secondary Users

- Playlist builders
- Music nerds who enjoy scene lineage
- Users who feel Spotify recommendations are repetitive
- Fans of indie, emo, post-hardcore, folk rock, pop-punk, shoegaze, alt-rock, or niche scenes
- Users who want artist-focused discovery rather than song-only recommendations

Primary Use Cases

Use Case	Description
Spotify Account Mode	The user signs in with Spotify. EchoFinder imports available listening data and helps the user choose which artists should act as legacy anchors.
Search by Artist	The user enters one or more artists manually, such as Manchester Orchestra, Thrice, and The Decemberists.
Search by Genre	The user enters a genre or subgenre, such as emotional indie rock, modern post-hardcore, literate indie rock, folk-influenced indie, or art rock.
Search by Scene	The user searches by a cultural or scene label, such as modern emo revival, indie rock storytellers, or progressive post-hardcore.
Hybrid Discovery	The user signs in with Spotify, then narrows the search manually by selected legacy artists, emergence window, genre, or scene.

Product Goals

- Help users discover newer artists based on older favorite artists, genres, or scenes.
- Prioritize artists who emerged within the last 0–5 years.
- Validate that recommended artists are currently active.
- Filter recommendations by practical success level.
- Explain why each artist fits.
- Support both Spotify login and manual search.
- Generate Spotify playlists from recommendations.
- Allow users to tune familiarity versus novelty.
- Capture feedback to improve future recommendations.
- Avoid dependence on restricted Spotify endpoints.

Non-Goals

- Replace Spotify as a listening platform.
- Guarantee perfect artist formation dates.
- Recommend only obscure artists or only mainstream artists.
- Treat popularity as the only measure of success.
- Require Spotify login for all functionality.
- Pretend genre tags are perfectly consistent.

- Build a full social network or industry analytics platform in the MVP.

Product Positioning

Element	Copy
One-sentence positioning	EchoFinder helps music fans discover active artists from the last five years who sound like modern echoes of the artists they already love.
Short marketing copy	Find the new bands carrying the sound you miss.
Tagline	Find the modern echo of the music you love.
App-store style copy	Connect Spotify or search by artist, genre, or scene. EchoFinder finds emerging artists who sound like your legacy favorites and explains why they fit.

Core UX Flow

Step 1: Choose Discovery Mode

- Connect Spotify
- Search by Artist
- Search by Genre
- Search by Scene

Step 2: Build or Confirm Legacy Anchors

In Spotify Account Mode, EchoFinder imports listening data and proposes likely legacy anchors. In Manual Mode, users enter artists, genres, scenes, moods, or songs directly.

Step 3: Configure Discovery Filters

Filter	Options
Emergence Window	Last 12 months; Last 2 years; Last 5 years; Last 10 years; Any age, but currently active
Activity	Released in last 12 months; Released in last 24 months; Touring or visibly active; Allow inactive artists
Success Tier	Hidden gems; Emerging; Growing; Established niche; Near mainstream
Similarity	Very close match; Balanced; Stretch discoveries
Discovery Source	My Spotify taste; Specific artist; Genre; Scene; Mood

Default settings: Last 5 years, active in last 24 months, emerging/growing/established niche, balanced similarity.

Step 4: Generate Echo Profile

Legacy Foundation:

- Manchester Orchestra
- Thrice

- The Decemberists

Sound DNA:

- emotionally intense indie rock
- post-hardcore dynamics
- literate storytelling
- folk-rock textures
- cathartic builds
- poetic or spiritual lyricism
- warm guitars
- dramatic arrangements

Step 5: Generate Modern Matches

- Artist name
- Echo Score
- Emergence window
- Activity status
- Success tier
- Genre/style tags
- Best legacy match
- Best starting tracks
- Why this fits
- Evidence used
- Open in Spotify
- Save
- Add to playlist
- More like this / Less like this

Echo Score

EchoFinder ranks candidates using a composite score called the Echo Score.

Component	Weight	Description
Style Similarity	30%	How closely the candidate matches the seed artist, genre, or taste profile.
Scene / Genre Lineage	20%	Whether the artist belongs to a modern continuation of the older sound.
Mood and Lyrical Fit	15%	Emotional and thematic overlap.
Current Activity	15%	Release, tour, press, playlist, or social activity.
Success / Traction	10%	Practical evidence that the artist has an active audience.
Recent Emergence	10%	Whether the artist fits the 0–5 year emergence scope.

Score Components

Style Similarity

- Genre tags
- Subgenre tags
- Artist descriptions
- Playlist contexts
- Music press language
- User feedback
- Release metadata
- Embeddings from artist bios or reviews

For Manchester Orchestra, Thrice, and The Decemberists, important style features include indie rock, post-hardcore, folk rock, art rock, emo-adjacent indie, progressive rock, literate songwriting, dynamic builds, emotional vocals, and narrative lyrics.

Scene / Genre Lineage

Legacy Artist	Possible Lineage
Manchester Orchestra	Emotional indie rock, emo-adjacent art rock, cathartic alternative
Thrice	Melodic post-hardcore, progressive post-hardcore, spiritual heavy rock
The Decemberists	Literate indie folk, narrative songwriting, chamber folk, folk rock

Mood and Lyrical Fit

- confessional
- cathartic
- literary
- melancholic
- spiritual
- existential
- narrative
- dramatic
- romantic
- anxious
- sincere
- wistful
- mythic
- working-class

Current Activity

- Release in last 12 months
- Release in last 24 months
- Multiple releases in last 3 years
- Touring or festival activity
- Recent playlist additions
- Recent press coverage

- Active label page or artist website
- Not marked as ended or disbanded

Success / Traction

- Playlist appearances
- Last.fm listeners or scrobbles
- Press mentions
- Tour listings
- Festival placements
- Label affiliation
- YouTube activity
- Bandcamp support
- Social activity
- Release frequency
- User saves and follows inside EchoFinder

Tier	Meaning
Hidden Gem	Very small but real footprint
Emerging	Newer artist with visible early activity
Growing	Clear playlist, listener, press, or scene traction
Established Niche	Known within a scene and actively followed
Near Mainstream	Broad visibility without necessarily being chart-dominant

Recent Emergence

Label	Meaning
High	First release or begin date clearly falls within 0–5 years
Medium	Earliest verified activity falls within 0–5 years
Low	Artist appears newly active, but origin data is incomplete
Breakout Recent	Artist is older but gained meaningful traction in last 0–5 years
Unknown	Not enough evidence; include only if user allows uncertain matches

Data Sources

Spotify

- OAuth login
- User top artists and tracks
- Saved tracks
- Followed artists
- Selected playlists
- Artist names, IDs, images, genres where available
- Albums, tracks, release dates
- Spotify links

- Playlist creation
- Saving tracks or artists
- Playback handoff

MusicBrainz

- Artist identity resolution
- Artist begin dates when available
- Release groups
- Release history
- Country or area
- Aliases
- Tags
- External IDs
- Disambiguation
- Ended status when available

Last.fm

- Artist tags
- Similar artists
- Listening behavior signals
- Scrobble/listener activity
- User listening history if connected
- Taste profile enrichment

Optional External Sources

- Bandcamp
- Songkick or Bandsintown-style tour sources
- YouTube
- music blogs
- festival lineups
- label websites
- artist websites
- public playlist databases
- review aggregators
- social platforms where terms allow

Candidate Discovery Logic

Spotify Account Mode

Extract legacy anchors

↓

Build genre and mood profile

↓

Identify related current scenes

↓

Generate candidate artists from tags, playlists, metadata, and similarity sources

↓

Filter for 0–5 year emergence



Validate activity



Score and rank

Manual Artist Mode

- Artist name
- Resolved artist ID
- Genre tags
- Artist descriptions
- Scene associations
- Release history

Genre / Scene Mode

- Expand the genre into related tags and subgenres
- Identify current candidate artists
- Filter by 0–5 year emergence
- Filter by activity and traction
- Rank by fit and confidence

Recommendation Output

Artist: Example Artist

Echo Score: 92%

Emergence: High confidence — first release 2022

Activity: Active — released music in 2025

Success Tier: Growing

Best Legacy Match: Manchester Orchestra + Thrice

Why this fits:

This artist shares emotional indie rock dynamics, cathartic vocals, and post-hardcore-adjacent arrangements while remaining current and active.

Production guidance: if an artist is older but still relevant, label them as Breakout Recent, Older Current Fit, Legacy Bridge, or Bridge Artist rather than misrepresenting them as newly emerged.

Core Screens

Screen	Purpose
Landing Page	Explains the value proposition and offers Connect Spotify, Search by Artist, Search by Genre, and Explore by Scene.
Spotify Import	Shows top artists, top tracks, saved artists, selected playlists, potential legacy anchors, listening era breakdown, and genre clusters.
Manual Search	Tabs for Artist, Genre, Scene, Mood, and Song.
Echo Profile	Displays Legacy Foundation, Sound DNA, Mood Profile, Scene Lineage, Excluded Sounds, and

	Discovery Settings.
Results	Recommendation cards, filters, sorting, compare mode, lineage preview, playlist builder, and feedback controls.
Artist Detail	Artist overview, Echo Score, emergence evidence, activity timeline, why-this-fits, tracks, tags, and Spotify actions.
Lineage	Shows how legacy favorites connect to bridge artists and modern echoes.
Playlist Builder	Creates playlists from high-confidence matches and allows auto-refresh settings.

Data Model

Model	Key Fields
User	user_id, spotify_user_id, email, display_name, created_at, updated_at
Artist	artist_id, spotify_artist_id, musicbrainz_artist_id, lastfm_artist_id, name, country, begin_date, first_release_date, first_verified_activity_date, breakout_date, active_status, genres, tags, images, spotify_url, external_urls
SearchSession	search_session_id, user_id, search_mode, query_text, emergence_window, activity_filter, success_filter, similarity_filter, novelty_filter, created_at
SeedArtist	seed_artist_id, search_session_id, artist_id, source, weight, excluded_flag, created_at
CandidateArtist	candidate_artist_id, search_session_id, artist_id, style_similarity_score, scene_lineage_score, mood_fit_score, activity_score, traction_score, recent_emergence_score, echo_score, success_tier, emergence_confidence_label, explanation_text
ArtistSignal	signal_id, artist_id, signal_type, signal_source, signal_value, signal_date, confidence, created_at
RecommendationFeedback	feedback_id, user_id, search_session_id, candidate_artist_id, action, reason, created_at

Technical Architecture

Layer	Recommended Stack / Responsibilities
Frontend	Next.js, React, Tailwind CSS, Spotify OAuth, responsive web app, mobile-first design system
Backend	Python FastAPI, PostgreSQL, Redis, Celery or RQ, SQLAlchemy or SQLModel
Data / Search	PostgreSQL, pgvector or Qdrant later, Redis cache, background enrichment workers
Backend Responsibilities	Spotify OAuth, API integrations, artist resolution, metadata enrichment, candidate generation, scoring, feedback storage, playlist creation,

MVP Scope

MVP Must Include

- Landing page with three entry paths
- Spotify OAuth
- Manual artist search
- Manual genre search
- Artist disambiguation
- Echo Profile generation
- Candidate artist generation
- 0–5 year emergence filter
- Current activity filter
- Success tier filter
- Echo Score ranking
- Explainable recommendation cards
- Spotify links
- Spotify playlist creation
- Thumbs-up / thumbs-down feedback

MVP Should Exclude

- Native mobile app
- Full social network
- Real-time concert alerts
- Complex user-to-user recommendation graph
- Heavy proprietary audio analysis
- Paid music-industry analytics integrations
- Full lyrics analysis unless licensing is solved
- Perfect formation-date verification

Version Roadmap

Version	Focus
Version 1: MVP	Spotify sign-in, manual search, 0–5 year emerging artist recommendations, explainable cards, playlist creation, basic feedback loop
Version 2: Scene Intelligence	Scene maps, genre revival timelines, modern successor graphs, local-scene clustering, familiarity vs novelty slider
Version 3: Live Music Layer	Local show recommendations, tour activity score, festival lineup discovery, artist alerts
Version 4: Community and Curator Layer	Public Echo Profiles, curator lists, discovery trails, scene experts, shared playlists

Risks and Mitigations

Risk	Mitigation
Spotify API restrictions	Use Spotify for authentication, catalog display, and playlist creation. Use MusicBrainz, Last.fm, metadata, tags, embeddings, release history, and user feedback for recommendation intelligence.
Popularity data limitations	Use multiple traction signals instead of one popularity number.
Artist emergence dates are messy	Track first release, first verified activity, and breakout date separately. Show confidence labels.
"Decently successful" is subjective	Use success tiers, allow threshold adjustment, and capture feedback such as too obscure or too popular.
Genre tags are inconsistent	Normalize tags into internal clusters and combine multiple metadata sources.
Recommendations may drift too old	Separate Modern Echoes from Bridge Artists and label older current-fit artists clearly.

Success Metrics

Metric Category	Examples
Activation	Spotify connection rate, manual search completion rate, seed artist selection rate, Echo Profile confirmation rate, first recommendation generated
Engagement	Recommendation cards opened, artist detail views, Spotify links clicked, artists saved, tracks saved, playlists created, repeat searches
Recommendation Quality	Thumbs-up rate, thumbs-down rate, already-known rate, wrong-genre rate, too-obscure rate, too-popular rate, not-recent-enough rate
Discovery Success	New artists discovered, recommended artists followed, recommended tracks saved, playlists retained, user rating of match quality

Example Scenario

User Input

Search by Artist:
Manchester Orchestra
Thrice
The Decemberists

Settings:
Emergence Window: Last 5 years
Activity: Active in last 24 months
Success Tier: Emerging to Established Niche
Similarity: Balanced

Generated Echo Profile

Legacy Foundation:
Manchester Orchestra
Thrice
The Decemberists

Sound DNA:
Emotional indie rock
Post-hardcore dynamics
Literate storytelling
Folk-rock textures
Cathartic builds
Poetic lyrics
Dramatic arrangements

Playlist Generated

Modern Echoes of Manchester Orchestra, Thrice & The Decemberists

Brand Identity Direction

Element	Direction
Brand Name	EchoFinder
Tagline	Find the modern echo of the music you love.
Personality	Nostalgic but current, dark-mode native, music-tech focused, premium but approachable, curated rather than chaotic
Palette	Near Black #0B0E14, Deep Slate #171B24, Electric Violet #4B3CF5, Signal Teal #19D2C2, Warm Coral #FF6B6B, Soft White #F2F1EE, Muted Gray #9CA3AF
UI Motifs	Waveform dividers, echo ripple icons, lineage graph nodes, gradient trails, artist cards, status badges, Sound DNA bars, playlist panels
Badges	Active, Growing, Best Match, New Since 2021, Breakout Recent, Emerging, Established Niche, Bridge Artist

Competitive Differentiation

EchoFinder combines legacy artist anchoring, manual and Spotify-based discovery, 0–5 year recent emergence filtering, current activity validation, success and traction tiers, scene and genre lineage, explainable recommendation reasoning, playlist creation, and user feedback tuning.

Most discovery tools offer broad personalization, similar artist graphs, genre exploration, or analytics. EchoFinder’s niche is the complete flow: older taste → current artists → recent emergence → active now → enough traction → explained clearly.

Final Product Definition

EchoFinder is a Spotify-centered discovery application that helps users find active artists from the last five years who sound like modern echoes of their favorite older artists, genres, and scenes. Users can connect Spotify for personalized taste analysis or search manually by artist, genre, scene, mood, or song. EchoFinder combines listening data, music metadata, tags, release history, activity signals, traction indicators, and explainable scoring to recommend current artists with the same musical DNA as the music users already love.

Core value: EchoFinder helps users discover new music without losing the sound that made them love music in the first place.

Appendix: MVP Pre-Work Checklist

Immediate next tasks before building the full UI:

6. Create Spotify Developer app.
7. Create Last.fm API key.
8. Write simple Spotify artist lookup script.
9. Write MusicBrainz artist/release lookup script.
10. Write Last.fm tag/similar-artist lookup script.
11. Create initial PostgreSQL schema draft.
12. Build Python prototype using Manchester Orchestra, Thrice, and The Decemberists.
13. Define first version of controlled tag/mood/scene taxonomy.
14. Create first Echo Score calculation in Python.
15. Manually review first 25 recommendations for quality.

Pre-work milestone: Before building the app UI, prove that the recommendation prototype can return at least 10 useful, explainable modern matches from a legacy artist set.